

Prototyping – GP3 – Parts I and II

What I want you to do

This is a two-part assignment in which you will create *prototypes* of your design solution. Think of prototypes like rough drafts; they can be *lower-fidelity* (really rough) to *higher-fidelity* (closer to the finished product). In Part I of this assignment, you'll create low-fidelity prototypes of two ideas, and then select one to build out in higher fidelity. As a class, we'll critique each group's higher-fidelity idea. Then, for Part II, you'll submit a new prototype, revised based on the feedback you received.

Why I want you to do it

This assignment is designed to simulate the rapid prototyping process that is typical of real-world design scenarios. Prototyping is inherent to human-centered design—and not to mention that it's resonant with agile project management more broadly. Prototyping has myriad advantages: it gives all stakeholders a clear picture of the benefits, risks and costs of a product idea; it allows you to show an early version of the product to your users to gather behavioral feedback; and it allows your team to adapt changes early, helping avoid *local maxima* in the solution space. In industry, parallel prototyping is a tried-and-true method for generating better solutions more quickly, and that's what we'll practice in this assignment. Additionally, the critique aspect of this assignment will give you a window into design culture and help you practice giving and getting feedback.

How to do it

This assignment is split into two parts. Part I will be submitted first, and then we'll have a critique session with the class. The following week, you'll submit Part II, which is a revised version of Part I. Along the way, you can use [Lyssna](#) to do small tests to aid in your decision-making. Lyssna offers many different ways for you to uncover more targeted usability issues (first click tests, design surveys, preference tests, five second memory tests, and more).

Suggested roles

- lo-fi prototyping 1
 - lo-fi prototyping 2
 - hi-fi prototyping
 - Lyssna researcher
 - writing up the process
 - editing/proofreading
 - project management
- Note that roles and team members may not line up one-to-one!

Part I

How to do Part I

Before you begin Part I, I'd like you to explore the [Lyssna](#) platform to get a sense of what is possible with it. We'll mostly be using it for GP4 to do usability testing (on Lyssna, they're called Prototype Tests), but you should have enough credits across your group to make some use of Lyssna here in GP3 if you'd like (more on that below).

In Part I, you'll create low-fidelity prototypes for two very different design ideas. The first stage in doing so is **brainstorming**. With your group, you should first come up with 10+ different ideas that you might pursue. The point at this stage is to come up with as many ideas as you can. *Quantity over quality*. Don't judge or criticize just yet. Instead, see if you can build on each other's ideas. Don't worry if it's good or bad, feasible or impossible. To guide your brainstorming session, go through these conversation-sparking questions with your group:

- *What would the product look like if...* it had to be a wearable? if it had to be an accessory for a self-driving car? if it had to be solar-powered? etc. (Oftentimes giving yourself constraints is a good way to discover unique ideas!)

- *What kind of conceptual model might your system have?* Think in terms of objects and actions. For instance, an email app has the objects of message, thread, person, etc.; and the actions of send, archive, etc. What objects and actions should be in your new system based on your design insights?
- *How might you bring your idea to life in different interface types?* The book talks about many interface types. Choose a few of them and talk through how your design might be using that kind of interface. What are the pros and cons? Here are the types of interfaces: Command line, Graphical, Multimedia, Virtual reality, Web, Mobile, Appliance, Voice, Pen, Touch, Gesture, Haptic, Multimodal, Shareable... (more details in the Sharp textbook!)
- *What metaphors might you use?* Many desktop apps use the metaphor of windows. (Even the idea of "desktop" is a metaphor!) On mobile, many apps nowadays are structured with the metaphor of cards. What would your idea look like in terms of cards or windows? What about other metaphors? Feel free to pose other idea-generating questions you find or that come to mind! There are no bad ideas at this stage. Even the zaniest idea may lead to a more realistic one later on.

Through your conversation, you should generate a **list of possible solutions** to explore for your design—that is, different types of forms of solutions that could address your problem statement. Then, **decide on two** that seem most viable. You could use Lyssna here to do a preference test, design survey, etc., to help you decide. Then it would be a data-driven decision!

At this stage, please be sure that the product/service idea you are considering is **interactive**. That means a user can do something, then something happens, then the user is presented with a result, and they can do something next. Your product doesn't need to be digital, but it should be interactive.

Now, you'll create **low-fidelity prototypes** of those two using paper or another low-fi tool (check out the Resources section on Blackboard for many suggestions). As we learned about in the textbook, lectures and in-class activities, these prototypes should:

- Meet the needs identified in your research insights
- Demonstrate that your product relates to the class design theme
- Clearly depict the key elements of the design (they needn't be fully polished)
- Include all screens/steps needed to complete the intended tasks

Next, you will **choose one** of these prototypes to develop further into a **high-fidelity prototype**. You could use Lyssna here to help you make this choice, and/or you can discuss the relative merits of each design idea as a group. At this point, you will create a higher-fidelity prototype for this design. Do this with prototyping software such as Figma (again, see the Resources section on Blackboard). Your prototype should:

- Be created using prototyping software
- Clearly depict the key elements of the design
- Include all screens/steps needed to complete the intended tasks
- Show a clear evolution from the lo-fi prototype—your hi-fi prototype should be at least semi-functional and more advanced in terms of the graphic design
- Be accessible to relevant users with disabilities and follow best practices for inclusive and universal design

You'll also be asked to give some rationale and reflection about your product (see the content requirements below).

What to turn in for Part I

Please submit a written report with images embedded (in PDF or Microsoft Word format), and your prototype files as separate documents. (Blackboard allows you to submit multiple files for a single assignment.)

Your **written report** should include the following elements. They don't have to be in sections exactly like this, but make sure all this content is there somewhere:

- Introduction (be concise)

- Brief description of the 10+ ideas you came up with initially (about a sentence each)
- Deeper description of the two low-fidelity prototypes you created, with images embedded for illustration (about 1/2 page each)
- Brief discussion of the relative merits and drawbacks of the two designs, and your rationale for choosing one over the other, making reference to concepts discussed in class (about 1/2 page)
- Description of the higher-fidelity prototype you made, emphasizing especially how it builds upon and differs from the low-fidelity version (about 1/2 page)
- Reflection on the merits of your final product choice with respect to accessibility, and what accessibility considerations you'll have to make as you flesh out your design in GP3 Part II through GP5. For this part, you can refer to any accessibility principles that came up in class, such as the POUR principles or the WCAG standards.
- Conclusion
- Also, if you used Lyssna along the way, be sure to explain where you used it, what tests you did, and what you learned from them.
- As an addendum, include a statement on individual roles and contributions, a sentence or two for each group member explaining what that person did for this assignment

As separate files, submit your **prototype files**. Depending on the software you used, these may be a Figma URL, photographs, PDF files, etc. The filetype is not so important. What is more important is that you explain and annotate each screen/display. The sequence of screens needs to be clear (number them); give each screen a title; and point to all the features and functionalities and briefly say what they are. A person should be able to inspect this file and understand what the user would be doing and thinking at each stage of the interaction.

The chief purpose of Part I is to create something that you can show to your classmates for critique and feedback. You'll use this feedback to make improvements to the interface and submit a revised prototype for Part II.

Guidelines

Here's what I'm looking for in this assignment:

Low-fidelity prototypes

- Meets the needs identified in your research insights
- Clearly and fully depicts the key elements of the design
- Adheres to design best practices
- The two designs present sufficiently distinct approaches to addressing the problem statement
- The two designs were chosen from a large set of possible contenders (10+)
- Consideration of the relative strengths and weaknesses of each design concept

High-fidelity prototype

- Created using prototyping software
- Clearly depicts the key elements of the design
- Includes all screens needed to complete the intended tasks
- Shows an evolution from the low-fidelity prototype
- Adheres to design best practices, including accessibility

Writing

- Clear thinking
- Convincing rationale
- Use of UX design concepts and principles

Part II

How to do Part II

The overall goal for Part II is to revise your prototype based on issues that you discovered after completing Part I. Hopefully, our in-class design critique session gave you a lot of food for thought. You can feel free to run (additional) Lyssna tests to answer further questions you have, but keep in mind that GP4 will focus on using Lyssna for usability research.

Additionally, you should do a **heuristic evaluation** of your prototype. For information on heuristic evaluation, see the textbook section “Inspections: Heuristic Evaluation and Walk-Throughs” (it’s section 16.2 in the 2019 edition). You can feel free to use Nielsen’s classic list of ten heuristics, or any of the other lists described in the text.

For a complete how-to on heuristic evaluation, [see this page](#). In brief, what you’ll be doing is this:

- Each individual team member runs through the prototype thoroughly several times, looking for violations of each heuristic. These will be kept in a list, and rated for severity: 1 indicates cosmetic problem, 2 indicates a minor usability problem, 3 indicates a major usability problem, and 4 is a show-stopping catastrophe.
- Then, the team gets together and everyone shares their results. You’ll discuss the different problems you found and come up with a master list. Deciding on whether each item is really a violation and coming to agreement on severity ratings will take some negotiation.
- Based on your master list of heuristic violations, come up with a list of changes you should make to the interface.

Now, take everything you learned and do the actual revisions to the prototype. You can use the existing prototype file as a baseline and build up from there.

What to turn in for Part II

- your group’s revised prototype
- a list of changes you made to your prototype and the source of that change (e.g., heuristic evaluation, critique session, in-group discussion, etc.)—this can just be a bulleted list
- the negotiated summary of your group’s heuristic evaluation
- each group member’s individual heuristic evaluation results